Dividing (2) by (1) and (3) by (2) we get

$$\frac{a}{x} = \frac{6}{n-1} \qquad (4)$$
$$\frac{a}{x} = \frac{9}{2(n-2)} \qquad (5)$$

From (4) and (5)

$$\frac{6}{n-1} = \frac{9}{2(n-2)}$$

Thus $n = 5$. Substituting $n = 5$ in (1) and (4) also dividing (1) by (4), we get

$$\frac{5x^4 a}{\frac{a}{x}} = \frac{240}{\frac{6}{4}}$$

Thus $5x^5 = 160$ and hence $x = 2$. Substituting in (4) we get $a = 3$.

Example 5.8 Expand $\left(2x - \frac{1}{2x}\right)^4$.

Solution:
We have

$$\begin{aligned}\left(2x - \frac{1}{2x}\right)^4 &= {}^4\mathrm{C}_0\,(2x)^4\left(-\frac{1}{2x}\right)^0 + {}^4\mathrm{C}_1\,(2x)^3\left(-\frac{1}{2x}\right)^1 + {}^4\mathrm{C}_2\,(2x)^2\left(-\frac{1}{2x}\right)^2 \\ &\quad + {}^4\mathrm{C}_3\,(2x)^1\left(-\frac{1}{2x}\right)^3 + {}^4\mathrm{C}_4\,(2x)^0\left(-\frac{1}{2x}\right)^4 \\ &= (2x)^4 - 4(2x)^3\left(\frac{1}{2x}\right) + 6(2x)^2\left(\frac{1}{2x}\right)^2 - 4(2x)\left(\frac{1}{2x}\right)^3 + \left(\frac{1}{2x}\right)^4 \\ &= 16x^4 - 16x^2 + 6 - \frac{1}{x^2} + \frac{1}{16x^4}\end{aligned}$$

Example 5.9 Expand $\left(x^2 + \sqrt{1-x^2}\right)^5 + \left(x^2 - \sqrt{1-x^2}\right)^5$.

Solution:
We have

$$\begin{aligned}\left(x^2 + \sqrt{1-x^2}\right)^5 &= {}^5\mathrm{C}_0\,(x^2)^5\left(\sqrt{1-x^2}\right)^0 + {}^5\mathrm{C}_1\,(x^2)^4\left(\sqrt{1-x^2}\right)^1 \\ &\quad + {}^5\mathrm{C}_2\,(x^2)^3\left(\sqrt{1-x^2}\right)^2 + {}^5\mathrm{C}_3\,(x^2)^2\left(\sqrt{1-x^2}\right)^3 \\ &\quad + {}^5\mathrm{C}_4\,(x^2)^1\left(\sqrt{1-x^2}\right)^4 + {}^5\mathrm{C}_5\,(x^2)^0\left(\sqrt{1-x^2}\right)^5 \\ &= x^{10} + 5x^8\sqrt{1-x^2} + 10x^6(1-x^2) + 10x^4(1-x^2)\sqrt{1-x^2} \\ &\quad + 5x^2(1-x^2)^2 + (1-x^2)^2(\sqrt{1-x^2})\end{aligned}$$